



Autonomous Clash Royale Artificial Intelligence

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Introduction

Clash Royale is a real-time strategy mobile game requiring fast decision making under uncertainty. This project builds a fully autonomous AI agent that plays the game by perceiving the screen, reasoning about game state, and executing actions without any game API access.

Novelty

Unlike most game AI research that relies on formal APIs or simulators, this system operates entirely from raw screen pixels on a live Android emulator. It combines computer vision (YOLO + template matching), real-time object tracking (SORT), and reinforcement learning (MaskablePPO) in a closed-loop pipeline without any API access or memory reading. Improvements over existing algorithms are shown in Fig. 3.

Methods

Each frame, two YOLO models detect troops and spells. A SORT tracker assigns persistent IDs, and team classification uses arena position to distinguish friendly from enemy units. Hand, elixir, and tower health are extracted via template matching and color sampling (Fig 2). A 260-float observation vector feeds a MaskablePPO agent, action masked by current elixir. The full data flow can be seen in Fig 1.

Results

The agent is able to identify all game state elements in real time, including troop positions, card hand contents, and tower health. It consistently plays cards at legal times, respects elixir costs via action masking, and targets both lanes based on observed threats. When testing, the AI won 1 out of 10 games against high skill human player(13.5k trophies). The model consistently damages enemy tower to 50%.

Impact & Future Work

This framework demonstrates that RL agents can operate on commercial mobile games without privileged access and establishes a reusable vision and tracking pipeline.

Future work includes training the agent across multiple different decks, developing the capability to play against a variety of opponent decks beyond the current fixed one and exploring finer-grained placement actions to improve spatial decision-making. Additionally, the more aspects of the game could added, including time and enemy hand tracking

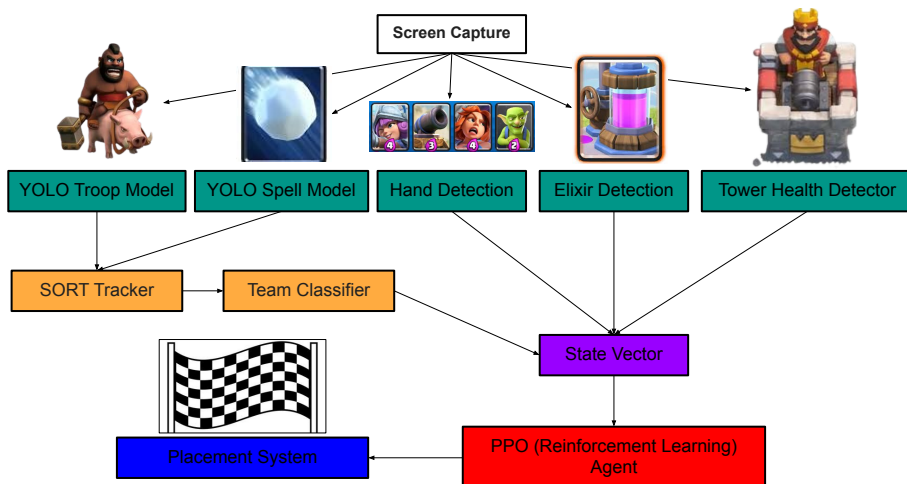


Fig 1: Data Flow

	Basic Bot	Ombarus	Junior Builds	My model
Elixir Tracking	No	Yes	Yes	Yes
Ally/enemy classification	No	No	No	Yes
Tower health detection	No	No	No	Yes
Structured game state	No	Partial	Partial	Yes
Reinforcement learning	No	No	Yes	Yes
Publicly reproducible	No	No	No	Yes

Fig 3: Comparison between my model and previous work



Fig 2: Areas of the game the model understands